Homework — week 1

Statistics in Physical Education and Sport · chapters 1–5 · submit in Teams before the next class (9–13 Nov 2026)

The problems come from the textbook instructor guide (Weir & Vincent, 2021, “Problems to Solve”). Solve them by hand first, then repeat in jamovi. The data are in the files STAT_DAS_N01_Homework_*.xlsx (one file per data set, one column X; jamovi opens only the first sheet). Submit one document: answers, the procedure, and jamovi output. Grade: 0–10 points.

Chapter 1 — concepts

1. Define measurement, data, statistics, and evaluation, and explain how they relate to one another.
2. What is the difference between a variable and a constant? Between an independent and a dependent variable? Give an example from sport.
3. List the four scales of measurement and give an example from sport for each. Explain why the example belongs to that scale.
4. Explain statistical inference. Define population, sample, random sample, and stratified random sample.
5. What is the difference between a parameter and a statistic?

Chapter 2 — 4 × 10-yard shuttle run (file STAT_DAS_N01_Homework_Shuttle.xlsx, seconds to the nearest tenth, N = 72)

6. Find the highest and the lowest score and compute the range.
7. Find the interval size (i). Note that the scores are recorded to the nearest tenth of a second.
8. Build the intervals and give each a frequency and a cumulative frequency. Make the lower limit of the lowest interval a multiple of 0.5. What is N?
9. What are the real limits of the interval 12.0–12.4?
10. Draw a histogram, a frequency polygon, and a cumulative frequency curve. Is the distribution roughly normal or skewed? What would probably happen to the polygon if more scores were added?
11. In jamovi, build the grouped distribution and the histogram and compare them with your hand result. Create the interval with a new variable: select X, Data → Transform, Create New Transform, formula FLOOR(($source + 0.05) / 0.5) * 0.5 = lower limit of the interval, Measure type Ordinal; then use Exploration → Descriptives → Frequency tables (Compute cannot create an ordinal variable).

Chapter 3 — percentiles

12. Tennis serve accuracy (file STAT_DAS_N01_Homework_Tennis_11.xlsx, N = 11): what is the percentile rank of a score of 7, and of 10? Which score is nearest to the 70th percentile, and which to the 45th?
13. The same test in three classes (file STAT_DAS_N01_Homework_Tennis_70.xlsx, N = 70): what is the percentile rank of a score of 16, and of 5? Which score best represents the 47th percentile, and which the 80th? Do it in jamovi as well (Frequency tables, Cumulative % column) and compare with your hand calculation.
14. Sit-ups in 1 minute, grouped distribution (table below, N = 83, by hand only): what are the percentile ranks of scores of 32, 39, and 55? Which scores best represent Q1, Q2, and Q3?

Interval	f	Cum. f
70–74	2	83

Interval	f	Cum. f
65–69	4	81
60–64	7	77
55–59	9	70
50–54	10	61
45–49	12	51
40–44	13	39
35–39	11	26
30–34	7	15
25–29	5	8
20–24	3	3

Chapter 4 — measures of central tendency

15. Compute the mode, the median, and the mean for set A (file STAT_DAS_N01_Homework_Set_A.xlsx, N = 15) and for set B (file STAT_DAS_N01_Homework_Set_B.xlsx, N = 60). Then do the same in jamovi (Descriptives: Mode, Median, Mean) and compare.

Chapter 5 — measures of variability

16. Sum of triceps and subscapular skinfolds in millimeters (file STAT_DAS_N01_Homework_Skinfolds.xlsx, N = 9): what is the standard deviation (SD) if the data are a whole population, and if they are a sample? What is the population variance?
17. Basketball shots made in 1 minute (file STAT_DAS_N01_Homework_Basketball.xlsx, a sample): in jamovi, compute N, the mean, the median, the mode, and the SD.

References

Weir, J. P., & Vincent, W. J. (2021). *Statistics in kinesiology* (5th ed.). Human Kinetics.

Weir, J. P., & Vincent, W. J. (2021). *Statistics in kinesiology: Instructor guide* (5th ed.). Human Kinetics.

The jamovi project. (2026). *jamovi* (Version 2.7) [Computer software]. https://www.jamovi.org

Use of artificial intelligence: the author led the work on this material from the initial idea to the finished content, designing it, setting the rules for writing and language use, and checking every part. Claude Code (Anthropic), an AI-based assistant, was used to write the programs that generate the slides, figures and documents, to draft text, and to check the language against the author's rules. The author is responsible for the content.